भारतीय प्रौद्योगिकी संस्थान जोधपुर
एनएच 65, सुरपुरा बाईपास रोड, करवड़, राजस्थान 342037

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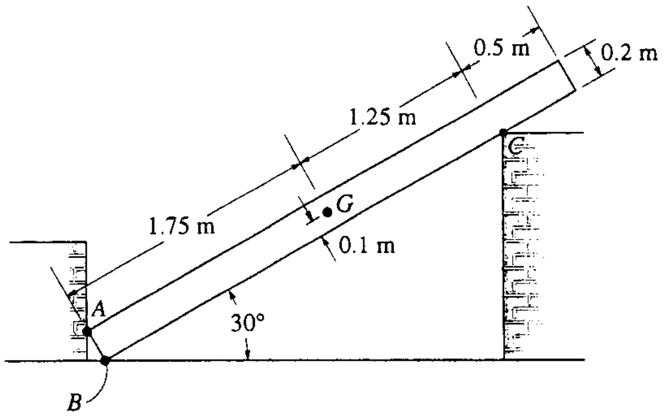
|| त्वं ज्ञानमयो विज्ञानमयोऽसि ||

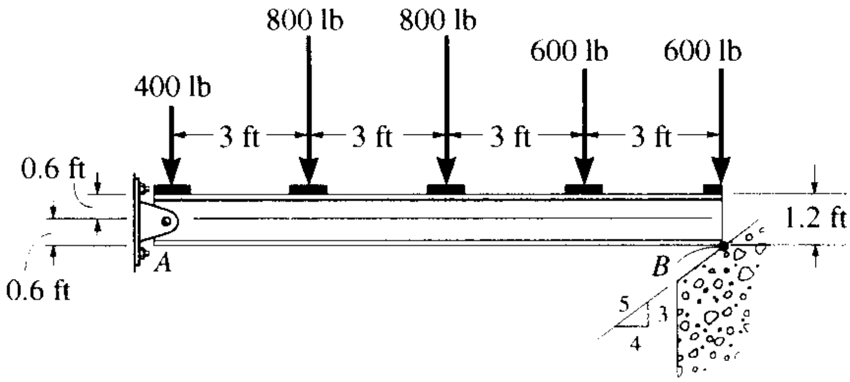
Course: Engineering Mechanics | **Code: MEL1010**

Date: 05/09/2024 | **Tutorial: 04**

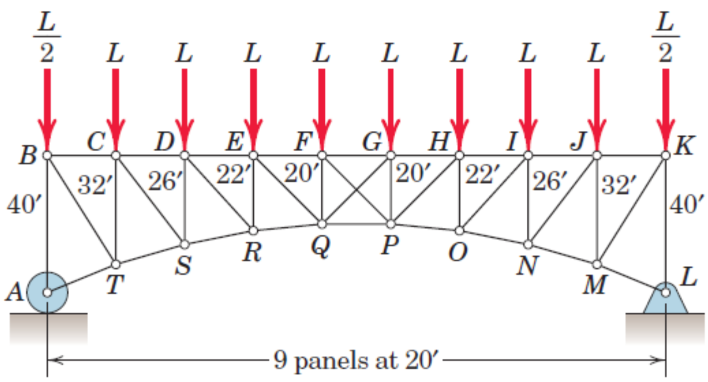
Question 01: A uniform bar has a mass of 100 kg and centre of mass at G. The supports A, B and C are smooth. Determine the reactions at the points of contact at A, B and C of the bar.



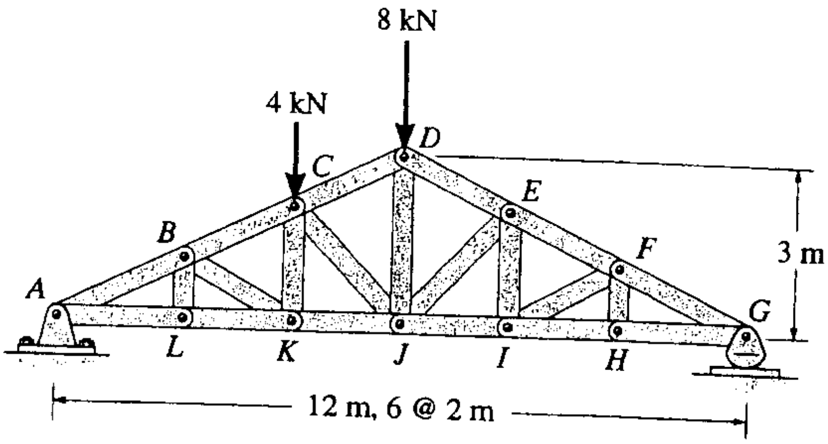
Question 02: Determine the support reactions in the following problem.



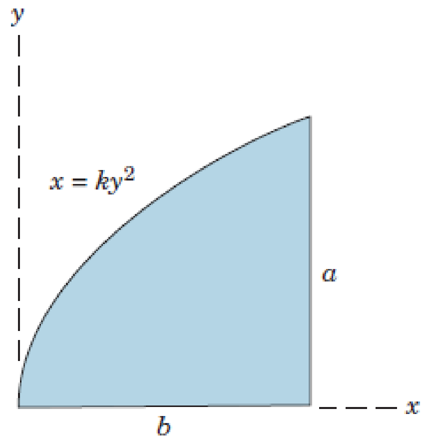
Question 03: Determine the force in member *HP* of the loaded truss. Members *FP* and *GQ* cross without touching and are incapable of supporting compression.



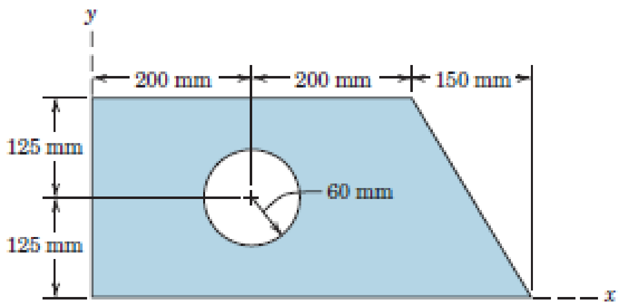
Question 04: The roof truss supports the vertical loading shown. Determine the force in members BC, CK and KJ and state if the members are in tension or compression.



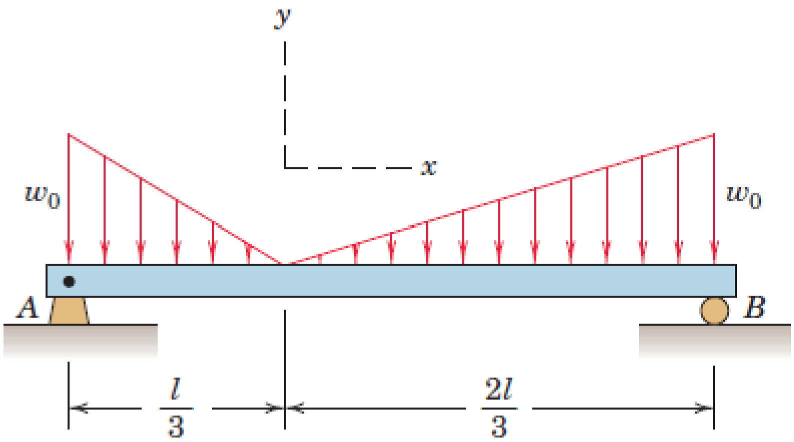
Question 05: Determine the coordinates of the centroid of the shaded area.



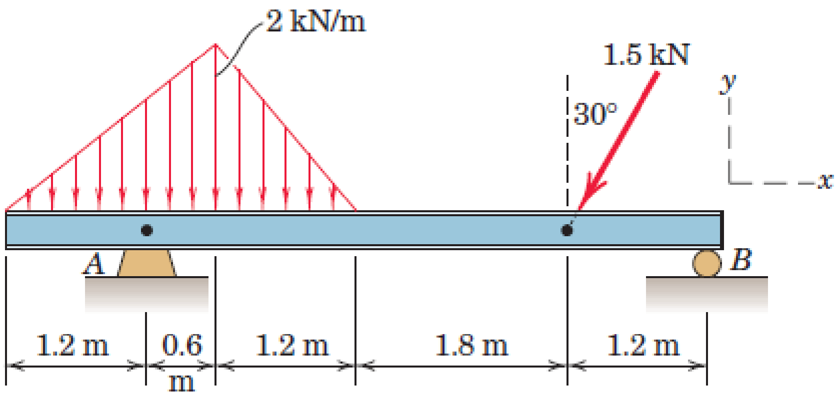
Question 06: Determine the coordinates of the centroid of the shaded area.



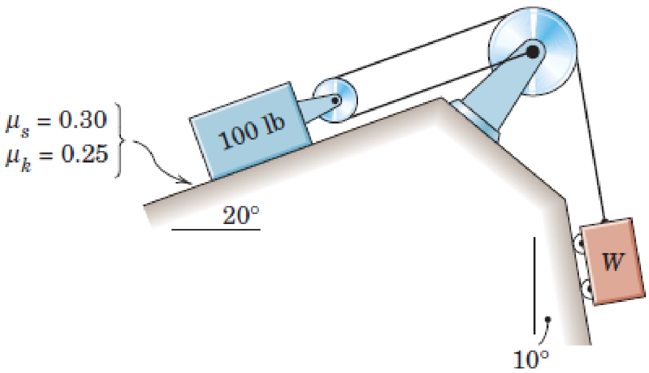
Question 07: Calculate the support reactions at A and B for the loaded beam.



Question 08: Determine the reactions at A and B for the beam subjected to a combination of distributed and point loads.



Question 09: Determine the range of weights W for which the 100-lb block is in equilibrium. All wheels and pulleys have negligible friction.



Question 10: The coefficient of static friction μs between the 100-lb body and the 15o wedge is 0.20. Determine the magnitude of the force P required to begin raising the 100-lb body if (a) rollers of negligible friction are present under the wedge, as illustrated, and (b) the rollers are removed and the coefficient of static friction μs = 0.20 applies at this surface as well.